

Master Math for JEE Main & JEE Advanced

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JEE Mains 2020 Jan Chapter wise Question Bank

Waves and Sound

Q1

Two sources of sound moving with same speed v and emitting frequency of 1400 Hz are moving such that one source s_1 is moving towards the observer and s_2 is moving away from observer. If observer hears beat frequency of 2 Hz. Then find the speed of source. Given

दो ध्वनि स्रोत एक समान चाल v से गतिमान है और 1400 Hz की आवृत्ति उत्सर्जित कर रहे है। एक स्रोत s_1 श्रोता की ओर गतिमान है तथा दूसरा स्रोत s_2 श्रोता से विपरीत दिशा में गतिमान है। यदि श्रोत 2 Hz की विस्पंद आवृत्ति सुनता है तो स्रोत की चाल होगी दिया गया है,

$$V_{\text{sound}} \gg V_{\text{Source}}$$

$$V_{\text{sound}} = 350 \text{ m/s}$$

$$(1) \frac{1}{4}$$

$$(2) 4$$

$$(3) 2$$

$$(4) \frac{1}{2}$$

7th Jan Evening

Sol

(1)



$$f_0 \left(\frac{C}{C - V} \right) - f_0 \left(\frac{C}{C + V} \right) = 2$$

$$V = \frac{1}{4} \text{ m/s}$$

Q2

An open organ pipe of length 1m contains an ideal gas whose density is twice the density of atmosphere at STP. Find the difference between fundamental and second harmonic frequencies if speed of sound in atmosphere is 300 m/s.

8th Jan Morning

Sol

105.75 Hz

$$V = \sqrt{\frac{B}{\rho}}$$

$$\frac{V_{\text{pipe}}}{V_{\text{air}}} = \frac{\sqrt{\frac{B}{2\rho}}}{\sqrt{\frac{B}{\rho}}} = \frac{1}{\sqrt{2}}$$

$$V_{\text{pipe}} = \frac{V_{\text{air}}}{\sqrt{2}}$$

$$f_n = \frac{(n+1)V_{\text{pipe}}}{2\ell}$$

$$f_1 - f_0 = \frac{V_{\text{pipe}}}{2\ell} = \frac{300}{2\sqrt{2}} = 105.75 \text{ Hz (If } \sqrt{2} = 1.41)$$

$$= 106.05 \text{ Hz (If } \sqrt{2} = 1.414)$$

Q3

Velocity of a wave in a wire is v when tension in it is $2.06 \times 10^4 \text{ N}$. Find value of tension in wire when velocity of wave become $\frac{v}{2}$.

किसी तार में गतिमान तरंग के वेग का मान v है, जब इसमें तनाव $2.06 \times 10^4 \text{ N}$ है। यदि तरंग वेग का मान $\frac{v}{2}$ हो जाये तो तार में तनाव ज्ञात किजिये।

- (1) $5.15 \times 10^3 \text{ N}$ (2) $8.24 \times 10^4 \text{ N}$ (3) 6×10^4 (4) $5.15 \times 10^4 \text{ N}$

8th Jan Evening

Sol

(1)

$$v \propto \sqrt{T}$$

$$\Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{v}{(v/2)} = \sqrt{\frac{2.06 \times 10^4}{T}}$$

$$\Rightarrow T = \frac{2.06 \times 10^4}{4} \text{ N} = 0.515 \times 10^4 \text{ N}$$

Q4

A string of mass per unit length $\mu = 6 \times 10^{-3} \text{ kg/m}$ is fixed at both ends under the tension 540 N. If the string is in resonance with consecutive frequencies 420 Hz and 490 Hz. Then find the length of the string?

एक रस्सी जिसका रेखीय द्रव्यमान घनत्व $\mu = 6 \times 10^{-3} \text{ kg/m}$ है, तथा तनाव 540 N है, दोनों सिरों पर बंधी हुई है। यदि रस्सी की दो क्रमागत आवृत्तियाँ 420 Hz तथा 490 Hz के साथ अनुनाद में है तो रस्सी की लम्बाई ज्ञात करो—

(1) 2.1 m

(2) 1.1 m

(3) 4.8 m

(4) 4.2 m

9th Jan Evening

Sol

(1)

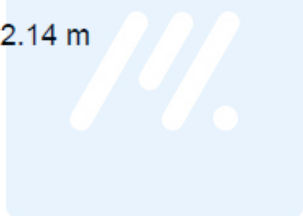
Fundamental frequency मूल आवृत्ति = $490 - 420 = 70 \text{ Hz}$

$$70 = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$$

$$\Rightarrow 70 = \frac{1}{2l} \sqrt{\frac{540}{6 \times 10^{-3}}}$$

$$\Rightarrow l = \frac{1}{2 \times 70} \sqrt{90 \times 10^3} = \frac{300}{140}$$

$$\Rightarrow l \approx 2.14 \text{ m}$$



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Wave Mechanics

Hints & Solutions

JEE Main 2020 Chapterwise

Physics

Q1

$$f = \frac{v}{2L}$$
$$\Rightarrow \frac{450}{300} = \sqrt{\frac{T_x}{T_z}}$$
$$\Rightarrow \frac{T_x}{T_z} = 2.25$$

Q2



$$V_P = (10 - 0.5)$$
$$= 9.5 \text{ m/s}$$

$$V_{PB} = 29.5 \text{ m/s}$$

Q3

$$m = A l \sigma \quad \therefore \mu = \left(\frac{m}{l} \right) = A \sigma$$

$$\text{Now, } v = \sqrt{\frac{T}{\mu}} \text{ and } \frac{T}{A} = Y (\text{strain})$$

$$\Rightarrow v = \sqrt{\frac{T}{\sigma A}} \Rightarrow \frac{T}{A} = 9 \times 10^{10} \times 4.9 \times 10^{-4}$$

$$\therefore v = \sqrt{\frac{9 \times 10^{10} \times 4.9 \times 10^{-4}}{9 \times 10^3}}$$

$$\Rightarrow v = 70 \text{ m/s}$$

$$\therefore f_0 = \frac{v}{2L} = \frac{70}{2 \times 1} = 35 \text{ Hz}$$

Q4

Wave Mechanics

Hints & Solutions

JEE Main 2020 Chapterwise

Physics

$$f = \frac{v}{\lambda}$$

$$\frac{V_1}{\lambda_1} = \frac{V_2}{\lambda_2}$$

$$\Rightarrow \lambda_2 = \lambda_1 \frac{V_2}{V_1}$$

$$= \lambda_1 \sqrt{\frac{T_2}{T_1}} = \lambda_1 \sqrt{\frac{8g}{2g}}$$

$$\Rightarrow \lambda_2 = 12\text{cm}$$

Q5

$$\left(\frac{\lambda}{2}\right) \times n = 1.5, \quad n \text{ is odd integer}$$

$$\lambda \times m = 5, \quad m \text{ is integer}$$

$$\Rightarrow \lambda \text{ can be } \frac{1}{1}, \frac{1}{3}, \frac{1}{5}$$

$$\text{as } \lambda = \frac{3}{n} \text{ and } \lambda = \frac{5}{m}$$

Q6

$$\frac{330+v}{330-v} \times 420 = 490$$

$$\Rightarrow \frac{330+v}{330-v} = \frac{7}{6}$$

$$\Rightarrow 13v = 330$$

$$\Rightarrow v = \frac{330}{13} = 25.38\text{m/s}$$

$$= 91\text{km/h}$$

Q7

$$\therefore \Delta P_0 = \frac{B\omega}{v} \times S_0$$

$$\Rightarrow S = \frac{\Delta P \times v}{B\omega} = \frac{\Delta P \times v}{\rho v^2 \times 2\pi v}$$

$$\Rightarrow S \propto \frac{\Delta P}{\rho v f}$$

$$\Rightarrow S = \frac{\Delta P}{\rho v f}$$

$$= \frac{10}{1 \times 300 \times 1000}$$

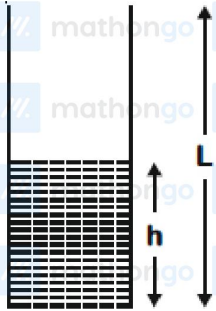
$$= \frac{1}{30} \text{mm}$$

$$\approx \frac{3}{100} \text{mm}$$

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Q8



$$L - h_1 = n \frac{\lambda}{2} + \frac{\lambda}{4}$$

$$L - h_2 = (n - 1) \frac{\lambda}{2} + \frac{\lambda}{4}$$

$$\Rightarrow \frac{\lambda}{2} = h_2 - h_1 = 24.5 - 17.0 = 7.5 \text{ cm}$$

$$\Rightarrow \lambda = 15 \text{ cm}$$

$$u = f\lambda$$

$$f = 2200 \text{ Hz}$$

Q9

$$\frac{345+u}{345-v} \times 440 = 480$$

$$\Rightarrow 11 \times 345 + 11v = 12 \times 345 - 12v$$

$$\Rightarrow v = \frac{345}{23} = 15 \text{ m/s}$$

$$= 54 \text{ km/h}$$

Q10

$$\Delta L, S_1 L - S_2 L = 2.5 - 2.0 = 0.5 \text{ m}$$

$$= \frac{\lambda}{2}$$

For adjacent maximum

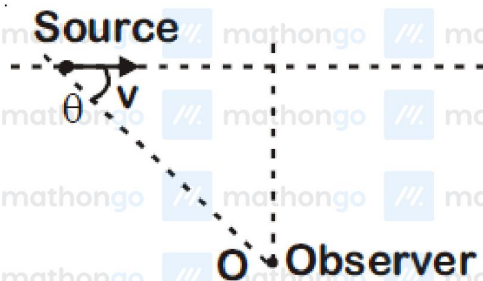
$$S_1 L - S_2 L = \lambda$$

$$S_1 L = \lambda + S_2 L$$

$$= 1 + 2$$

$$= 3 \text{ m}$$

Q11



While approaching

$$v = v_0 \left(\frac{c}{c - v \cos \theta} \right)$$

While receding

$$v = v_0 \left(\frac{c}{c + v \cos \theta} \right)$$

Q12

$$v = \sqrt{\frac{T}{\mu}}$$

$$T = \mu v^2$$

$$\frac{\mu v^2}{A} = Y \frac{\Delta \ell}{\ell}$$

$$\Delta \ell = \frac{\mu v^2 \ell}{AY}$$

after substituting value of μ , v , ℓ , A and Y we get

$$\Delta \ell = 0.03 \text{ mm}$$

Q13

$$f_0 \left(\frac{C}{C - V} \right) - f_0 \left(\frac{C}{C + V} \right) = 2$$


observer

$$V = \frac{1}{4} m/s$$

Q14

$$V = \sqrt{\frac{B}{\rho}}$$

$$\frac{V_{pipe}}{V_{air}} = \frac{\sqrt{\frac{B}{2\rho}}}{\sqrt{\frac{B}{\rho}}} = \frac{1}{\sqrt{2}}$$

$$V_{pipe} = \frac{V_{air}}{\sqrt{2}}$$

$$f_n = \frac{(n+1) V_{pipe}}{2\ell}$$

$$f_1 - f_0 = \frac{V_{pipe}}{2\ell} = \frac{300}{2\sqrt{2}}$$

$$= 105.75 \text{ Hz (If } \sqrt{2} = 1.41)$$

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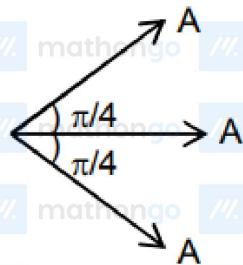
Q15

$$v \propto \sqrt{T}$$

$$\Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{T_1}{T_2}} \Rightarrow \frac{v}{(v/2)} = \sqrt{\frac{2.06 \times 10^4}{T}}$$

$$\Rightarrow T = \frac{2.06 \times 10^4}{4} N = 0.515 \times 10^4 N$$

Q16



$$A_{res} = (\sqrt{2} + 1) A$$

$$I_{res} = (\sqrt{2} + 1)^2 I_0$$

$$= (3 + 2\sqrt{2}) I_0 = 5.8 I_0$$

Q17

Fundamental frequency = $490 - 420 = 70 \text{ Hz}$

$$70 = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$$

$$\Rightarrow 70 = \frac{1}{2\ell} \sqrt{\frac{540}{6 \times 10^{-3}}}$$

$$\Rightarrow \ell = \frac{1}{2 \times 70} \sqrt{90 \times 10^3} = \frac{300}{140}$$

$$= \ell \approx 2.14 \text{ m}$$